Karnataka Residential Educational Institutions Society

ICT Level 3

Teaching Guide

Level: Level 3

Class: Classes 8, 9, 10

Duration: June - January (8 months)

Period: 40 minutes

ICT Teaching Resources Portal

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Curriculum Overview

ICT Level 3 covers advanced topics for Classes 8 through 10. The same textbook is used across all three classes, with differentiated instruction tailored to the developmental stage and academic expectations of each class.

Themes Covered

Chapter	Periods	Key Topics
Ch 1: Advanced Programming Concepts	12	Functions, Arrays, String operations, Nested loops
Ch 2: Data Structures	12	Lists, Dictionaries, Tuples
Ch 3: Database Fundamentals	12	Database concepts, SQL basics
Ch 4: Web Technologies	12	HTML basics, CSS introduction
Ch 5: Algorithms and Computational Thinking	12	Algorithm design, Pseudocode
Ch 6: Emerging Technologies	12	AI basics, IoT, Cloud computing

Differentiated Pedagogy by Class

Class 8 (Foundation)

Class 8 students require heavy Concrete-Phase-Abstract (C-P-A) scaffolding and guided practice. Instruction should rely on visual aids, step-by-step demonstrations, and structured worksheets. Peer collaboration is encouraged through pair programming and group tasks. Concepts are introduced with real-world analogies before transitioning to abstract representations.

Class 9 (Application)

Class 9 students benefit from a balanced approach between guided and independent learning. They are expected to apply previously learned concepts to new contexts with decreasing levels of teacher scaffolding. Emphasis is placed on debugging skills, logical reasoning, and collaborative problem-solving. Students begin to work on mini-projects with defined parameters.

Class 10 (Mastery)

Class 10 students focus on independent problem-solving and board exam preparation. Instruction emphasizes revision, application of concepts under timed conditions, and analysis of past examination papers. Students are expected to work with minimal scaffolding and demonstrate mastery through capstone projects and self-directed inquiry.

5E Model Implementation

Phase	Implementation Strategy
Engage	Activate prior knowledge through demonstrations, thought-provoking questions, or short videos related to the topic. Present a real-world problem that motivates the lesson.
Explore	Students engage in hands-on activities such as coding exercises, database queries, or web page construction. Teacher facilitates but does not direct.
Explain	Teacher-led discussion clarifies key concepts, introduces terminology, and formalises rules observed during exploration. Students articulate their understanding.
Elaborate	Students apply learning to new scenarios, extend existing programs, or tackle extension tasks. Emphasis on transfer of knowledge.
Evaluate	Formative assessment through observation, questioning, peer review, and exit tickets. Summative assessment through chapter-end exercises and projects.

Lesson Plan Structure (40 Minutes)

Time	Phase	Activity
0-5 min	Engage	Hook activity: demonstration, question, or quick recall quiz
5-15 min	Explore	Hands-on coding or activity worksheet; students work individually or in pairs
15-25 min	Explain	Teacher clarifies concepts, introduces terminology, addresses misconceptions
25-35 min	Elaborate	Extension task or application problem; independent or group work
35-40 min	Evaluate	Exit ticket, quick quiz, or self-reflection prompt

Assessment Framework

Assessment Type	Weightage	Components
Formative	40 marks	Class participation, Worksheets, Homework, Quizzes, Projects
Summative	60 marks	Mid-term exam, End-of-term exam, Practical assessment
Total	280 marks	Continuous assessment across three terms; final grade derived from cumulative scores

Resources Required

- Computers with Python installed (minimum one per pair of students)
- Textbook: ICT Level 3 (as prescribed by the state board)
- Printed worksheets and activity sheets for each chapter
- Projector and screen for demonstrations
- Access to online coding environments as backup (e.g., Replit, Trinket)
- Chart paper, markers, and sticky notes for C-P-A activities
- Database software (SQLite or equivalent for Chapter 3)
- Web browsers with developer tools enabled (for Chapter 4)

Dormitory Tasks

Dormitory tasks are assigned for evening self-study in the boarding facility. These tasks are designed to reinforce the day's learning and promote independent practice.

- **Code tracing exercises**: Students trace through given code snippets and predict output without running the program.
- **Debugging worksheets**: Intentionally flawed programs are provided for students to identify and correct errors.
- **Concept mapping**: Students create visual maps connecting related concepts from the chapter.
- **Reading assignments**: Selected sections of the textbook are assigned for pre-reading before the next lesson.
- **Mini-projects**: Short guided projects (e.g., building a calculator, creating a simple web page) that consolidate multiple concepts.

Capstone Projects

Capstone projects are assigned at the end of each term. They require students to integrate knowledge from multiple chapters and demonstrate cumulative understanding.

Project Guidelines

- Projects must address a real-world problem or scenario relevant to the student's context.
- Students may work individually or in groups of up to three.

- Each project requires a written component documenting the design process, algorithm, and reflections.
- Projects are presented to the class in the final week of the term.

Sample Project Topics

- **Library Management System**: A console-based program using functions, lists, and basic database operations to manage book records.
- **Personal Portfolio Website**: A multi-page website using HTML and CSS showcasing student work and interests.
- **Traffic Signal Simulator**: A Python program using loops and conditional logic to simulate traffic light sequences.
- **Data Analysis Report**: Students collect survey data, store it in a database, and generate a summary report using SQL queries.

Unplugged Activities Library

Activity 1: Sorting Algorithm Race (Ch 2 - Data Structures)

Duration: 20 minutes | **Materials:** Numbered cards (1-20)

Steps:

1. Give each group 20 shuffled number cards
2. Task: Sort them in ascending order using ONLY these rules:
   - You can only look at 2 cards at a time
   - You can swap 2 cards (but count each swap)
3. Try: Bubble sort vs. Selection sort. Which is faster?
4. Discuss: Why does algorithm efficiency matter?

CT Skill: Algorithm design, pattern recognition, evaluation

Activity 2: Database on Paper (Ch 3 - Databases)

Duration: 20 minutes | **Materials:** Graph paper, pencil

Steps:

1. Create a "Student" table with columns: Roll No, Name, Class, Marks
2. Fill in 10 rows of data
3. Answer queries: "Who scored above 80?", "How many students are in Class 8?"
4. Discuss: How is this different from scrolling through a list?

CT Skill: Abstraction, data organization

Activity 3: HTML on Paper (Ch 4 - Web Technologies)

Duration: 15 minutes | **Materials:** Paper, pencil

Steps:

1. Write the HTML for a simple page: heading, paragraph, image, link
2. Draw what the page would look like in a browser
3. Trade with another group and "render" their HTML mentally
4. Discuss: What does the browser actually do?

CT Skill: Abstraction, notation systems

Activity 4: Flowchart Relay (Ch 5 - Algorithms)

Duration: 20 minutes | **Materials:** Flowchart symbol cards

Steps:
1. Give each group a set of flowchart symbols (start, process, decision, end)
2. Task: Create a flowchart for "What to eat for lunch?"
3. Include at least 2 decision points
4. Trade and follow another group's flowchart exactly
5. Discuss: How do computers make decisions?

CT Skill: Algorithm design, logical reasoning

Activity 5: Ethical Dilemma Discussion (Ch 6 - Emerging Tech)
Duration: 20 minutes | **Materials:** Scenario cards

Steps:
1. Present scenarios: "Should a self-driving car save the passenger or a pedestrian?"
2. "Should AI be allowed to grade your exams?"
3. Students debate in groups and present their reasoning
4. Discuss: Who is responsible when technology makes mistakes?

CT Skill: Evaluation, ethical reasoning

Cross-Curricular Integration Map

Mathematics Connections
- **Ch 1:** Time complexity uses mathematical notation (Big-O)
- **Ch 2:** Arrays use indexing (coordinate geometry)
- **Ch 3:** Database normalization uses set theory
- **Ch 4:** CSS uses coordinate systems (box model)
- **Ch 5:** Algorithm analysis uses logarithms and exponents
- **Ch 6:** AI uses probability and statistics

Science Connections
- **Ch 1:** Exception handling models error correction in biological systems
- **Ch 2:** Data structures mirror natural organizations (tree structures in biology)
- **Ch 3:** Database relationships model scientific classification
- **Ch 4:** Web protocols use electromagnetic signals (Physics)
- **Ch 5:** Algorithms model natural processes (photosynthesis, circulation)
- **Ch 6:** Emerging tech intersects with all sciences

Language Connections
- **Ch 1:** Technical documentation and code commenting
- **Ch 2:** Data description and analysis writing
- **Ch 3:** SQL queries are structured language
- **Ch 4:** Web content requires clear, concise writing
- **Ch 5:** Algorithm descriptions require precise language
- **Ch 6:** Ethics essays and persuasive writing

Progressive Programming Pathway

Block-Based to Text-Based Transition
The Level 3 curriculum bridges visual programming (Scratch) to text-based programming (Python).

Class 8: Foundation

- **Term 1:** Scratch advanced features (clones, custom blocks, lists)
- **Term 2:** Python Turtle Graphics (visual transition)
- **Term 3:** Python basics (variables, loops, conditionals)

Class 9: Application

- **Term 1:** Python functions and file handling
- **Term 2:** Data structures in Python (lists, dictionaries)
- **Term 3:** Database operations with Python

Class 10: Mastery

- **Term 1:** Full Python projects + web development basics
- **Term 2:** Integration projects combining all skills
- **Term 3:** Board exam preparation with previous year papers

Transition Strategy

1. Start with Scratch blocks that map directly to Python:
   - `repeat 10` ⟶ `for i in range(10):`
   - `if on edge bounce` ⟶ `while True:`
   - `set x to 0` ⟶ `x = 0`
2. Use Python Turtle for visual output (immediate feedback)
3. Gradually introduce text-based debugging
4. Pair Scratch projects with equivalent Python projects

Digital Citizenship Module

Class 8-10: Advanced Digital Citizenship

Lesson: AI Ethics and Bias

- What is algorithmic bias? Show examples (facial recognition, hiring algorithms)
- Discuss: Can a computer be racist? Sexist?
- Rule: "AI reflects the data it's trained on - and data reflects human bias"

Lesson: Data Privacy

- Read the terms of service of a popular app (simplified version)
- What data does it collect? Who can see it?
- Create a "Privacy Checklist" for evaluating apps

Lesson: Online Reputation

- Google yourself or your school. What do potential employers/colleges see?
- Build a positive digital presence: portfolio, blog, GitHub
- Rule: "Your online presence is your extended resume"

Lesson: Cyber Law in India

- Overview of IT Act 2000 and its key provisions
- What is illegal online? (hacking, cyberbullying, identity theft)
- Know your rights and responsibilities as a digital citizen